

What the program is:

The program will be a chess learning tool. It will have the board in the center with the number and letters as grid references and on the right hand side will be buttons to switch side to play on that will flip the board, another button to reset the board to its original state and a button that when you click it will let you draw arrows on the board from the first square you click to the second square you click. and finally at the bottom right there will be a choice box that has all the available board states to pick from that can be loaded in by selecting it from the drop down menu.

How the program will be used:

This program will be used in teaching environments during lessons in order to teach children how to play chess and work through the moves and improve their skills. The teacher would share their screen of the program and pick the board state they want and then talk the students through the positions on the board and then ask and demonstrate how the position should be played on both sides by using the switch buttons and arrows to show the next moves from both sides and using the coordinates on the sides for the children to tell the teacher where the pieces are and should go.

How the program will work:

The program would work by first drawing a blank board of objects that would be a grid and then adding the coordinates position on the left and down side of the board and adding the buttons as well. then the user would pick which board state they want to load into via the drop down menu which then the text file would be opened and the board state read added to a list where each piece in the file will be their full name like pawn or knight and then their x and y coordinates so a line in the file could be (Knight A 3) showing that the knight piece is at position A3 and we can separate the colour of the pieces by if the first letter is a capital letter or not. if it is a capital it is a white piece. After the file is read the board would be loaded on and each coordinate has a tile with the same coordinate so the name of the piece would be stored in that tile as well as the colour so when it comes to drawing the piece it would ask what it is in that tile and then it would use that and string and check it with a list and when it is found draw it at the tiles x and y coordinates.

When the switch button is clicked the board would be flipped so the only differences in the code is that the for loop would go in reverse order for the y axis and then the board will be redrawn flipped for the other colour to be at the bottom and by flipping that all the pieces will be flipped as that is how the pieces are drawn.

the reset button would work by re-reading the file and redoing the loading process again so we would need a whole function or class just for that which is fine as that makes it abstract.

the arrow button would work by when you click the first square it would remember and then when you click the second square an arrow will be drawn using the coordinates during this the arrow button will turn red and after the arrow is drawn the red will go away and the arrow button will be deselected. I think I will do it as 64 buttons to make it easier than tracking the mouse position and

doing maths. this would also be easier as i could store the piece and data for the button in it which makes storing that info easier.

to choies the diffrent board stats there will be a menu on the top left that will have a drop down menu with the buttons upload or save. and the map will be stored using FEN notation (Forsyth–Edwards Notation) which will also store the positions. When saving the file it you will input a name to save it and it will sort the menu by date and show the name.

There will be a second menu on the left that has a drop down menu that will have a button for the piece and a second to say what colour you want the pieces and then you click on the tile that you want the piece. This is how you can create the files to be stored and open all from the program.

The 3 chalaging bits:

1.

I think the most challaging bits will be arrow drawing due to the math required and the thickness of the arrow to make it look good enough without fully hiding the pieces bellow it but if we make the arrows somewhat transparent it could work without hideing the pieces.

2.

I also think the second hardest thing would be flipping the board arround as i feel a lot of things could go wrong just by drawing the board with just the pieces fliped as we have lots of things reliing on that board. maybe if we do it as a variable we could use to toggel to flip that inverse all the need functions it would work smothly.

3.

I think i will find it hard to find all the files stored at the place and how to add them to the menu but i think i could be solved if you used a wild card like \*.txt to get all text files in that directory.

Picture:

